

Educational Manual for ISP v2.0

Planning and evaluating intervention studies in human processes

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Default problem family

This manual assumes interventions in human processes such as learning, training, usability, workflow, or decision support. It is not a clinical-trial manual for medication, surgery, or safety endpoints. It is strongest when the researcher needs practical guidance for sample-size planning or achieved-result interpretation in non-clinical intervention studies.

Intervention Sample Planner contributors

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1 What the Software Now Covers

ISP v2.0 was originally a two-group calculator. Version 2.0 extends the software in three directions:

- different **research paths**;
- recommended-value checking with explicit override;
- an **inverse workflow** for evaluating achieved results.

Path or mode	When it applies	What the software does
Two independent groups	Intervention and control are different people.	Plans or evaluates a between-group comparison.
Pre-test/post-test with control group	The same participants are measured before and after in both groups.	Uses an approximate gain-sensitive design with a pre/post correlation input.
One-group pre-test/post-test	There is no control group and the same participants are measured twice.	Plans or evaluates a paired change design, with explicit warnings about causal limits.
Plan required sample	Data have not yet been collected.	Estimates valid, assigned, and invited sample sizes.
Evaluate achieved result	Data or pilot results already exist.	Estimates approximate achieved power and p-value for the observed result.

Why the first wizard question changed

In version 2.0, the first question is no longer the outcome type. It is the research path. This matches the real logic of research planning: the same variable, such as effect size, means different things in different designs.

2 Choosing the Research Path

2.1 Two independent groups

Use this path when one group receives the intervention and another group does not. The central claim is about a difference between groups.

Worked scenario

A researcher in educational games wants to verify whether using the game Uno helps children learn the concepts of greater-than and less-than. The intervention group plays Uno with guided prompts and then receives a short lesson. The control group receives only the short lesson. Both groups complete a final test.

Why this path fits

The intervention is not defined as change inside the same participant over time. The main claim is whether the intervention group differs from the control group at the end of the activity sequence.

2.2 Pre-test/post-test with control group

Use this path when both groups are measured before and after. This is useful when baseline differences matter and when the researcher wants the pre-test to improve precision.

Worked scenario

A researcher in educational games wants a stronger design than post-test only. All children complete a pre-test on greater-than and less-than. The intervention group then plays Uno and receives a short lesson. The control group receives only the lesson. Both groups complete the same post-test.

Why this path fits

The software treats the pre-test as a precision-improving measurement and asks for the expected correlation between pre-test and post-test. Higher correlation usually means a more efficient design.

For readers who want more statistics: What the pre/post correlation does

When pre-test and post-test are moderately or strongly correlated, baseline information helps reduce unexplained variability. The version 2.0 implementation uses an approximate ANCOVA-style precision gain based on the factor $(1 - r^2)$, where r is the pre/post correlation.

2.3 One-group pre-test/post-test

Use this path when the same participants are measured before and after and there is no control group.

Worked scenario

A researcher wants an early pilot before attempting a controlled trial. The same children complete a pre-test, participate in a guided Uno session, and then complete a post-test. Final adherence is defined as completing both tests.

Why this path fits

The design is centered on change within the same participants rather than comparison against a separate group.

Important limit: What this path cannot establish strongly

Without a control group, observed improvement may reflect practice effects, history, maturation, or ordinary instruction rather than the intervention itself. This path is useful for pilots, feasibility work, and early signal detection, but not as strong causal evidence.

3 Recommended Ranges

Version 2.0 introduces recommended or typical ranges. The software checks these ranges and allows the user to continue only when a specific override box is selected for a field that falls outside the recommended range.

Variable	Recommended range	Typical values	Justification
alpha	0.01 to 0.10	0.05, 0.01	Outside this range, the evidential standard becomes uncommon and should be justified.
power	0.80 to 0.95	0.80, 0.90	Lower power weakens detection; higher power may become infeasible.
allocation_ratio	0.5 to 2.0	1.0	Balanced groups are usually efficient.
effect_size_d	0.10 to 1.20	0.20, 0.50, 0.80	Tiny effects can demand very large samples; huge effects should not be assumed casually.
pre_post_correlation	0.30 to 0.80	0.50, 0.60	Common range for many educational and usability measures.
response_rate	0.40 to 1.00	0.60, 0.80, 0.90	Low values make recruitment fragile.
completion_rate	0.70 to 1.00	0.85, 0.90, 0.95	In repeated-measure designs, completion means finishing all required waves.
usable_data_rate	0.80 to 1.00	0.90, 0.95, 0.98	Lower values often indicate a data-quality issue, not just a sample-size issue.
extra_buffer_rate	0.00 to 0.20	0.00, 0.05, 0.10	Small buffers are common; large buffers may indicate weak operational certainty.
intraclass_correlation	0.00 to 0.20	0.01, 0.05, 0.10	Even small ICC values can matter when clusters are large.

4 Understanding Effect Size Better

4.1 Why effect size is difficult

Effect size is difficult because it must connect statistics to a real decision. The researcher should not ask only, “What can I afford to detect?” The better question is, “What is the smallest effect that would matter enough to justify adopting the intervention?”

4.2 Effect size in a post-test-only two-group study

Suppose a post-test has a pooled standard deviation of 10 points. If a 5-point improvement would justify using the game-based intervention, then:

$$d = \frac{5}{10} = 0.5.$$

This is a traditional medium effect in Cohen’s language (Cohen 1988), but what matters most is the practical meaning of the 5-point gain.

4.3 Effect size in pre-test/post-test with control

In this design, the important effect is often the difference in gain. Imagine that both groups start at similar pre-test scores. If the intervention group improves enough after Uno and the control group improves less after lesson only, the researcher wants the study to detect that meaningful difference in change.

The entered `effect_size_d` should therefore reflect the smallest meaningful standardized difference in gain.

4.4 Effect size in one-group pre-test/post-test

In a one-group pre/post design, the effect is the standardized change in the same participants. This can be useful for early pilots and feasibility checks.

Important limit: Interpretation warning

A one-group pre/post effect size can describe how much change was observed, but not by itself why that change happened.

4.5 Effect size in opinion or usability studies

Sometimes the researcher has only a post-use questionnaire. In that case, the same logic still applies. If a 0.4-point difference on a 1-to-5 satisfaction scale would justify changing a game tutorial, and the expected pooled standard deviation is 0.8, then:

$$d = \frac{0.4}{0.8} = 0.5.$$

5 Run Type: Planning or Evaluating

5.1 Planning the sample

The planning workflow estimates:

- initial valid analyzable cases;
- valid cases after design correction;
- participants who must start;
- participants who must be invited.

This separation is especially important in pre-test/post-test studies because a participant should not count as complete unless the required measurement points are complete.

5.2 Evaluating the achieved result

The inverse workflow asks the user for the achieved sample and the observed effect. It then estimates an approximate:

- z statistic;
- p-value;
- achieved power for the observed effect.

Worked inverse scenario

A one-group Uno pilot collected 28 children who completed both tests. The observed standardized change was 0.35. The researcher wants to know what that implies.

Why this matters

If the desired effect was not found, the researcher should still report the effect actually observed and what the achieved sample was able to show about it.

6 Worked Examples

6.1 Uno with control, post-test comparison

A researcher in educational games wants to verify whether the use of Uno helps children understand greater-than and less-than. The intervention group plays Uno with guided prompts and then receives a short lesson. The control group receives only the lesson. The main comparison is the final test.

Recommended wizard path:

- research path: two independent groups;
- run type: plan required sample;
- outcome: continuous;
- effect size: for example, $d = 0.5$ if a 5-point gain on a 10-point spread would justify adopting the intervention.

6.2 Uno with pre-test/post-test and control

A researcher wants a stronger design. Both groups complete a pre-test. The intervention group plays Uno and receives a short lesson. The control group receives the lesson only. Both groups complete the same post-test. The researcher expects a moderate pre/post correlation, such as 0.60, and wants to detect a meaningful difference in gain.

Recommended wizard path:

- research path: pre-test/post-test with control group;
- run type: plan required sample;
- outcome: continuous;
- effect size: meaningful standardized difference in gain;
- pre/post correlation: use pilot data when possible, otherwise a reasoned value such as 0.50 or 0.60.

6.3 Uno with one group only

A researcher cannot yet recruit a comparison group and wants a pilot. The same children complete a pre-test, play Uno in a guided session, and complete a post-test.

Recommended wizard path:

- research path: one-group pre-test/post-test;
- run type: plan required sample;

- outcome: continuous;
- completion rate: define completion as finishing both tests.

7 Suggestions Tab

The new suggestions tab is a practical judgment layer. It is designed to translate methodological concerns into direct software feedback.

It can warn about:

- a no-control-group design;
- low response, completion, or usable-data expectations;
- cluster inflation;
- out-of-range values that the user deliberately overrode;
- achieved results that remain underpowered.

8 When Another Methodology Is Better

Situation	Why ISP v2.0 is not enough	Consider instead
Binary pre/post without a simple parallel comparison	The current implementation does not model paired binary change directly.	McNemar-based or specialized paired binary methods.
Many repeated waves over time	The design is longitudinal, not single pre/post.	Mixed models or longitudinal power analysis.
Interview-centered evidence	Mean/proportion power formulas do not define qualitative adequacy.	Information power or saturation logic (Malterud et al. 2016).
Clinical safety or medication	Ethical and regulatory constraints are much stronger.	Clinical-trial planning (Julious 2010).
Strong causal claims without control	Time and practice effects remain serious threats.	Add a control group or use a stronger quasi-experimental design (Shadish et al. 2002).

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